

A-05-
B-V, iv
C-7

38 + 2
55.5

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
FIRST SEMESTER 2024 - 2025

Civil Engineering Materials (Comprehensive Examination)

Course No: CE F230

02-12-2024 (Monday)

Duration: 180 Mins. (9 am - noon)

Max. Marks: 52 (Part A) + 28 (Part B) = 80

PART A (Closed book in-built: 52 Marks; No time limit)

Write neat, clean, and to the point answers

Name: MANAS CHOUDHARY	Exam Room no: 2201	Invigilator sign: A
ID: 2023A2PS1342P	Student signature: Manas	Marks obtained:

A. Answer the following questions

1. 4 sieves (19mm, 9.5mm, 4.75mm, 2.36 mm) were used for sieve analysis of aggregates and the cumulative % passing was (70%, 30%, 10%, and 0%) respectively. Determine the fineness modulus. [2]

19mm → 70%
9.5mm → 30%
4.75mm → 10%
2.36mm → 0%

$$FM = \frac{70 + 30 + 10 + 0}{100} = \frac{110}{100} = 1.1$$

⇒ FM = 1.1

2. What is plywood and what are its advantages over sawn lumber? [4]

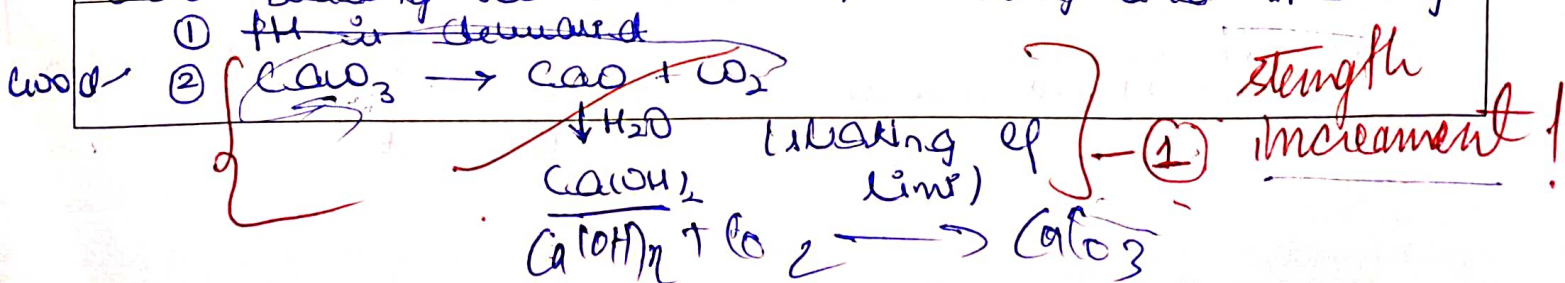
→ In plywood certain number of timber slabs/veners are placed so that it is balanced about its central core. In this the distribution is uniform.

→ ① Its properties along the length of the plywood is same about the width of the plywood.

② Due to its cross-laminated structure, it is resistant to splits.

- Bad 3. Mention good and bad things about carbonation in concrete. [4]

Depassivation → carbonation decreases the pH, destroying the protective thin film of iron oxide on steel reinforcement bars leading to corrosion, cracking and spalling.



(4)

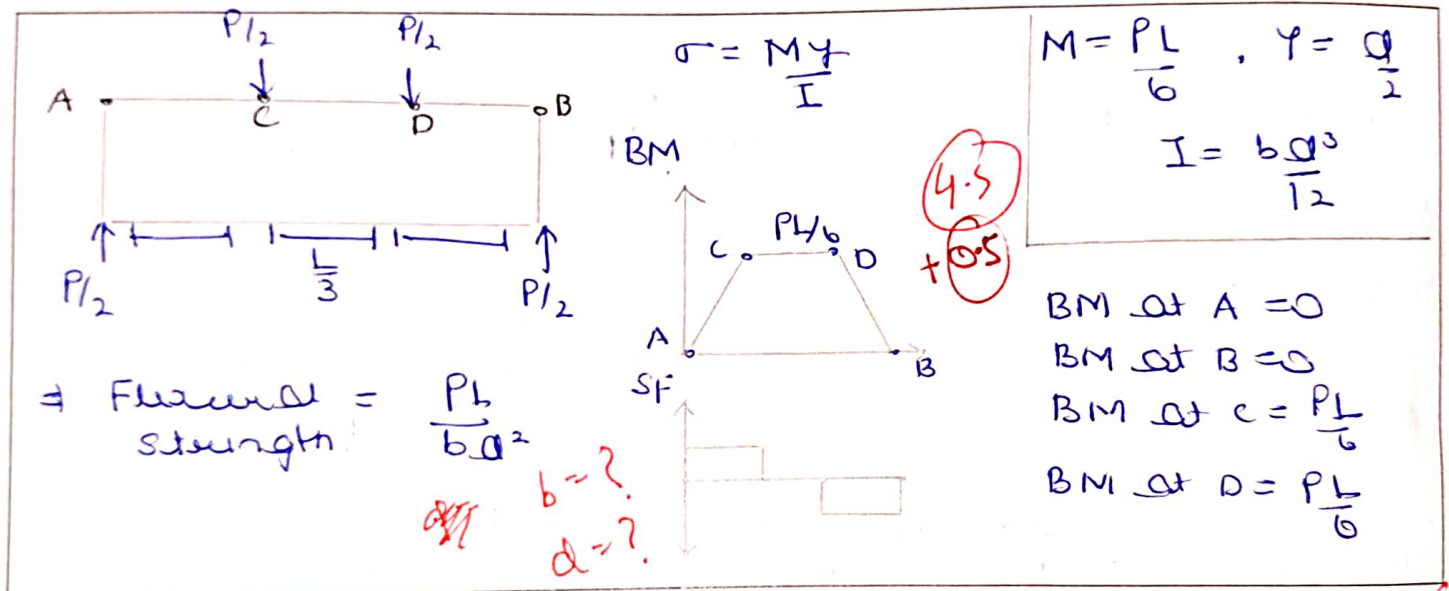
4. Remark on durability of high strength concrete against fire and advantages of high strength concrete. [4]

→ HSC is poor resistance to fire because when exposed to heat, water turns into steam which cannot escape out due to fire & dense pores. Due to which internal pressure increases, results in sudden cracks and blowing out.

→ ① cross section size can be reduced or for the same reinforcement (column) cross section the amount of steel reinforcement bar can be reduced.

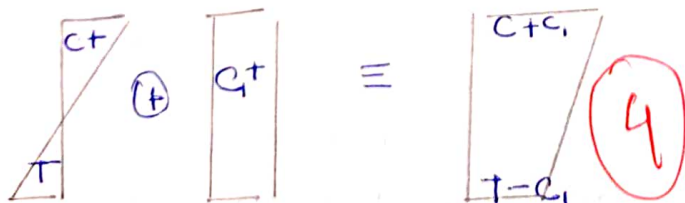
② In bridge, using HSC can reduce the number of beams.

5. Derive the formula for flexural strength in 4-point bending test. (Assume: P = max. total load, L = span length, b = specimen width, d = specimen depth). [5]



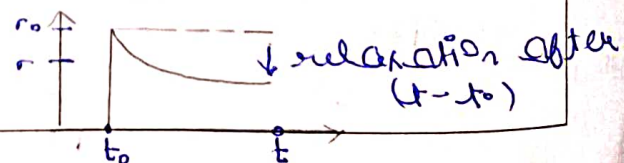
6. What is prestressing and how creep in concrete leads to loss of prestress. [5]

→ In Prestressing, we are intentionally applying tensile force to the concrete, that will counteract with the tensile force developed during the load gain.



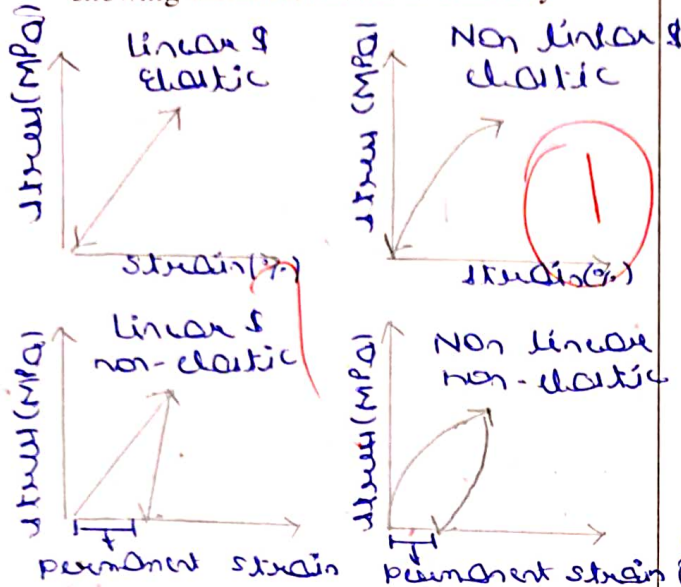
Creep → develops when there is increase in strain over time for constant stress

In relaxation, stress is gradually decreased keeping strain constant
 ⇒ leads to loss of prestress in creep

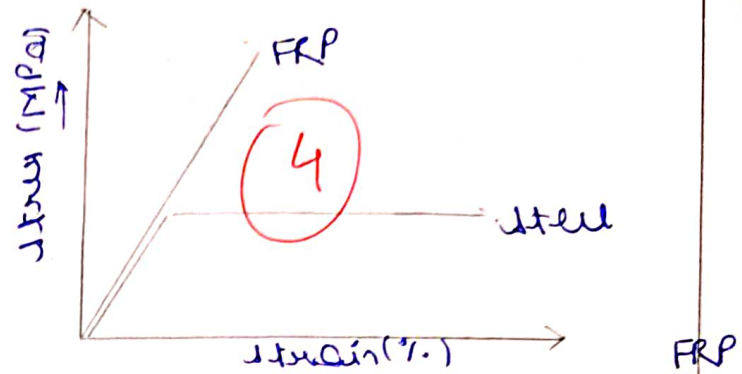


B. Plot the following [20]:

(i) Typical stress-strain response of concrete showing various modulus of elasticity



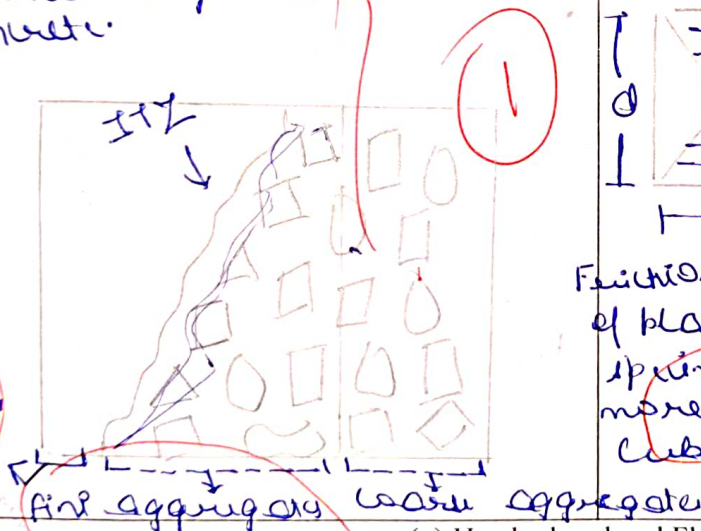
(ii) Comparative graph showing stress-strain response of FRP vs Steel in tension



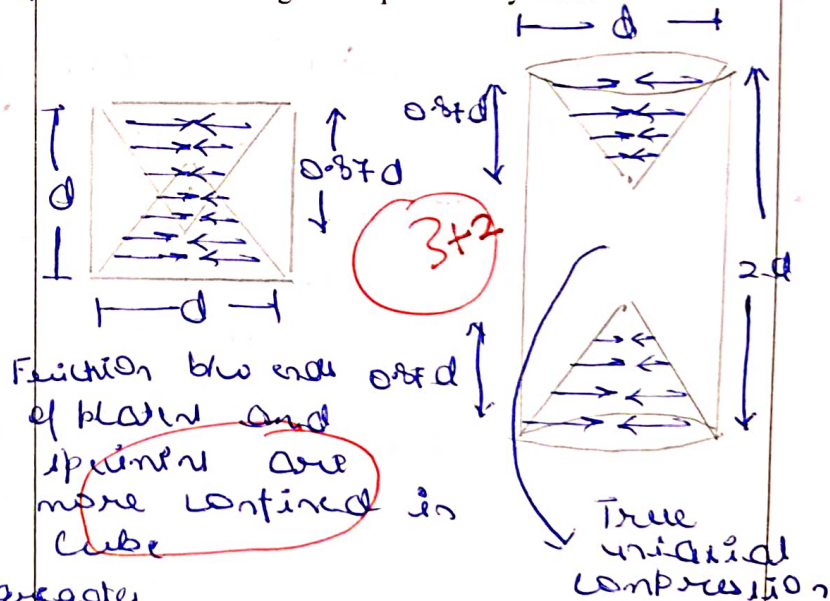
- ① high ultimate strength
- ② No yielding point
- ③ linear elastic behaviour to failure
- ④ comparable modulus

(iii) Microscopic view of ITZ

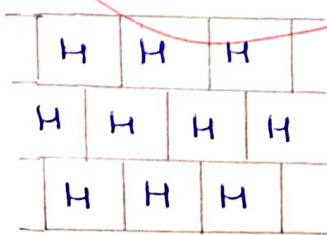
ITZ is interfacial transition zone which explains the permeability behaviour in concrete.



(iv) Confinement effect in cube leading to higher strength compared to cylinder

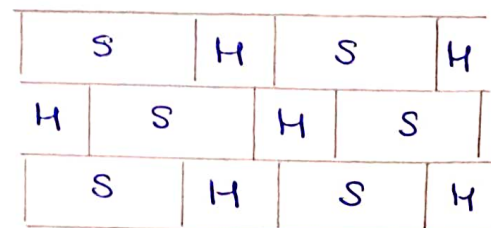


(v) Header bond and Flemish bond



S → stretch bond

H → header bond



Flemish Bond

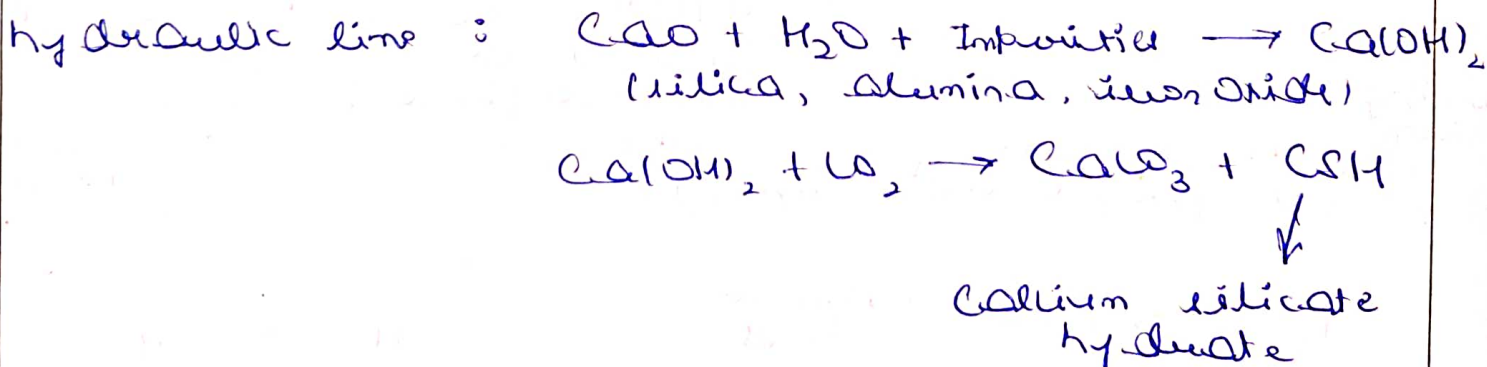
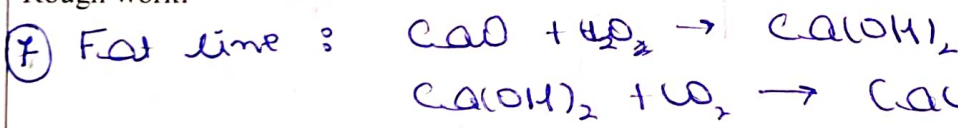
header bond (Quoin is placed at corner)

(4+1)

C. Answer the following questions [8]:

1. Cement required for rapid concrete repair work should be rich in $C_2S/C_3S/C_4AF$ C_3S
2. Cement particles are angular in shape whereas fly ash particles are spherical. So, if you replace cement with fly ash in concrete the workability would increase, and the early age strength would decrease, tendency of developing thermal cracks would decrease, and cost of concrete would decrease.
3. Fibers in the fiber reinforced concrete are used to increase post cracking ductility (toughness)
4. Among Fe 250, Fe 415 and Fe 500 which is generally the most ductile? Fe 500
5. The size of an Indian standard brick including mortar is 20 x 10 x 10 cm
6. The variation of strength between different directions can be attributed to anisotropic behaviour (coherent bonds in longitudinal & van der Waals wall, hydrogen bonds in transverse)
7. Hardening mechanism in case of fat lime is _____ while in case of hydraulic lime is CSH is extra produced
8. A cutback asphalt is a combination of asphalt cement and bitumen
9. To subject asphalt to hardening conditions approximating those that occur in normal HMA batch plant operations Test on Oven Dry TROP test is performed.
10. Name a safety test on bitumen Consistency / Penetration

Rough work:



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ID:	Student signature:	Marks obtained:

PART A (CLOSED BOOK - 52 Marks), Max. time: 2 hours

A. Answer the following questions

1. 4 sieves (19mm, 9.5mm, 4.75mm, 2.36 mm) were used for sieve analysis of aggregates and the cumulative % passing was (60%, 30%, 10%, and 0%) respectively. Determine the fineness modulus. [2]

19	60	40	$\frac{700}{100} = 7.0$
9.5	30	70	
4.75	10	90	
2.36	0	100	
1.18	0	100	
0.6	0	100	
0.3	0	100	
0.15	0	100	

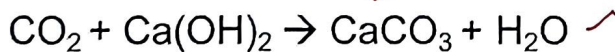
2. What is plywood and what are its advantages over sawn lumber? [4]

Plywood usually contains an uneven number of veneers so that the properties are 'balanced' about the central veneer or core. Thin veneers are glued together in layers (or plies), with the grains of the successive plies at right angles to each other. The principal advantages compared to sawn lumber are as follows:

- Because of its cross-laminated structure, plywood is very split resistant.
- Its properties along the length of the sheet are similar to its properties across the width.
- Because knots extend only through 1 ply, they are not particularly detrimental to strength.
- Shrinkage and swelling are minimized, because the tendency of an individual ply to swell or shrink perpendicular to the grain is restrained by the resistance to movement of the adjacent plies at right angles to the ply in question.

3. Explain good and bad things about carbonation in concrete. [4]

Conversion of CH to strength providing CaCO₃



Reduction in pH (to ~9 from 12.6-13.5 initially) which can cause passivation of steel rebar, leading to corrosion

Rough work:

4. Remark on durability of high strength concrete against fire and advantages of high strength concrete. [4]

poor resistance to fire because the very low permeability of high performance does not allow the egress of steam formed from water in the hydrated cement paste

- ☐ Column sections can be reduced in size or, for the same cross-section, the amount of steel reinforcement can be reduced. In tall buildings, there is an economic advantage because of an increased floor area for rental
- ☐ In bridges, the use of high strengths can reduce the number of beams.

5. Derive the formula for flexural strength in 4-point bending test. (Assume: P = max. total load, L = span length, b = specimen width, d = specimen depth). [5]

Use this formula, Bending stress = My/I ,

Find out maximum bending moment M for 3-point bending, use $y_{\max} = d/2$, use $I = bd^3/12$

Get flexural strength = PL/bd^2

6. What is prestressing and how creep in concrete leads to loss of prestress. [5]

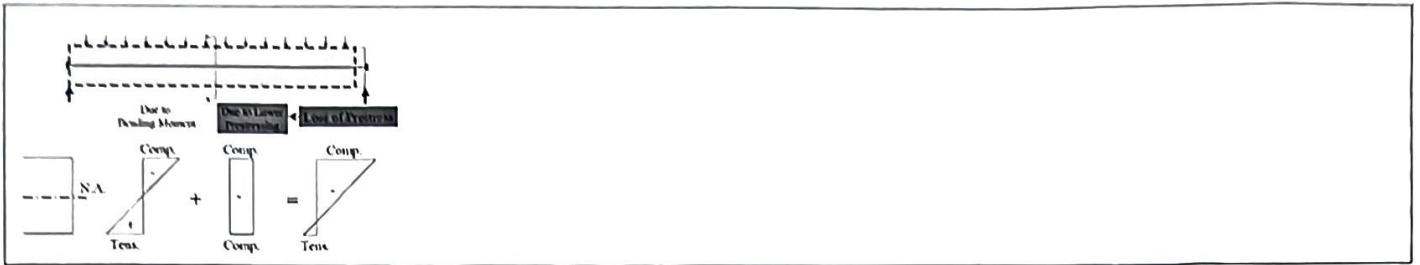
Concrete is weak in tension and strong in compression. This is a weak point of concrete that results in early flexural cracks mainly in flexural members like beams and slabs.

To prevent this, the concrete is induced with compressive stress deliberately (prestressing) and this stress counteracts with the tensile stress the structure is subjected to during service condition. Hence the chances of flexural cracks are reduced.

Due to creep, prestressed concrete has a tendency to shorten under the compressive load.

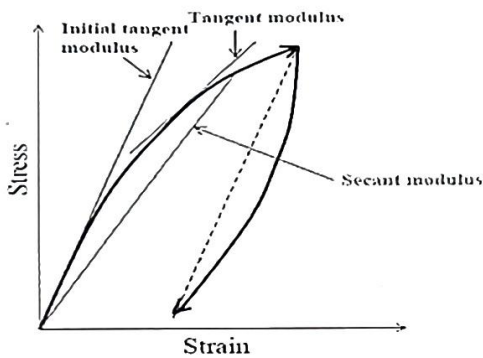
However, the prestressed concrete is in a constant strain condition.

Hence, it leads to relaxation of the prestressing forces and thus there is a loss of prestress.



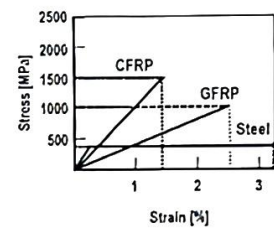
B. Plot the following [20]:

(i) Typical stress-strain response of concrete showing various modulus of elasticity

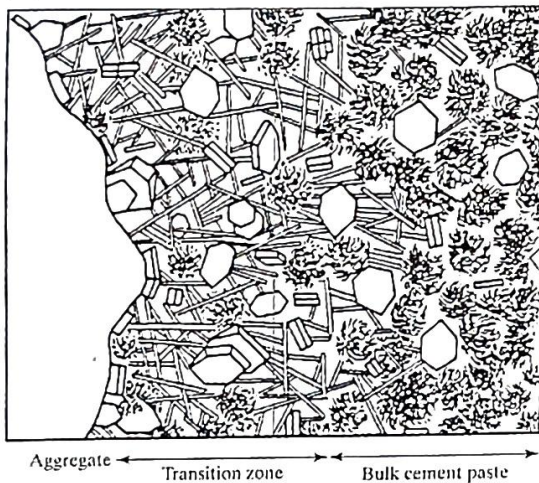


(ii) Comparative graph showing stress-strain response of FRP vs Steel in tension

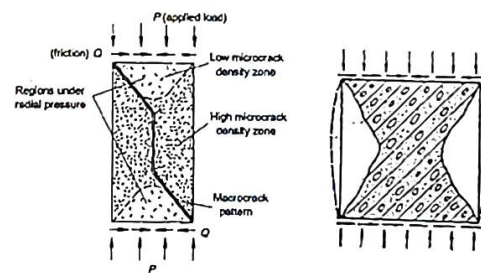
- FRP properties (in general versus steel):
 - Linear elastic behaviour to failure
 - No yielding
 - Higher ultimate strength
 - Lower strain at failure
 - Comparable modulus (carbon FRP)



(iii) Microscopic view of ITZ

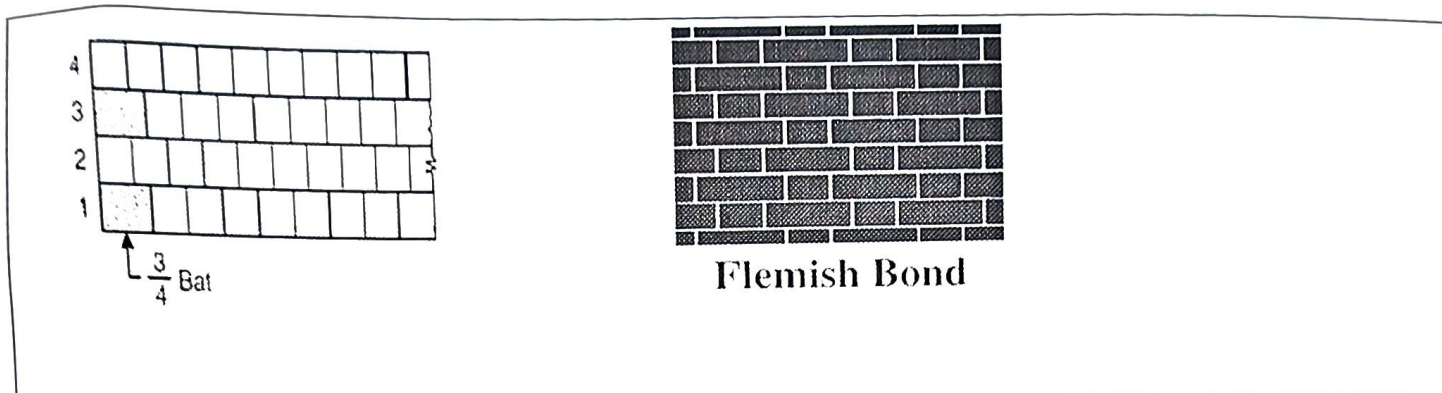


(iv) Confinement effect in cube leading to higher strength compared to cylinder



The confining pressure is greatest right at the specimen end and gradually dies out at a distance about $(\sqrt{3})d/2 \approx 0.87d$. For standard cylinder with $l/d = 2$, a small central portion of the cylinder is in true uniaxial compression.

(v) Header bond and Flemish bond



C. Answer the following questions [8]:

1. Cement required for rapid concrete repair work should be rich in ~~C₂S/C₃S/C₄AF~~ C₃S
2. Cement particles are angular in shape whereas fly ash particles are spherical. So, if you replace cement with fly ash in concrete the workability would improve / increase, and the early age strength would reduce / decrease, tendency of developing thermal cracks would reduce / decrease and cost of concrete would reduce / decrease.
3. Fibers in the fiber reinforced concrete are used to improve the post-cracking ductility
4. Among Fe 250, Fe 415 and Fe 500 which is generally the most ductile? Fe 250
5. The size of an Indian standard brick including mortar is (20 ~~mm~~ X 10 X 10) cm.
6. The variation of strength between different directions can be attributed to anisotropic nature of wood cells. ^{wood.}
7. Hardening mechanism in case of fat lime is carbonation only. ($\text{CaO} + \text{CO}_2 \rightarrow \text{CaCO}_3$) while in case of hydraulic lime is carbonates + silicates. ^{hydration of} (Fast)
8. A cutback asphalt is a combination of asphalt cement and petroleum solvent/ oil distillates Diesel oil / Kerosene
9. To subject asphalt to hardening conditions approximating those that occur in normal HMA batch plant operations TFOT/RTFOT test is performed. → THIN FILM OVEN TEST
10. Name a safety test on bitumen Flash and Fire point test / Flash Point Test / Fire Point test

***** Collect PART B *****

Space for rough work:

Final Solⁿ

- Chemical admixture : Superplasticizer confirming to IS 9103, Solid content = 40%
- Concrete compressive strength testing data for 10 cubes (values in MPa): 55, 58, 56, 54, 54, 48, 54, 56, 55, 59
- Coarse aggregate: 500g sample in SSD condition weighted 300g when immersed in water.
- Specific gravity of fine aggregate and SP are 2.60, and 1.20 respectively
- Water absorption of both coarse aggregate and fine aggregates are 1.5 %
- Moisture content of Coarse aggregate is 1.0 % and Fine aggregate is 2.0 %
- Sieve analysis results:

Coarse aggregate

Sieve size (mm)	Cum % passing
40	100
20	95
10	30
4.75	5
2.36	0

Fine aggregate

Sieve size (mm)	Cum % passing
10	100
4.75	95
2.36	90
1.18	80
0.6	70
0.3	14
0.15	5

In context of the above problem, answer the following (Only the final answer will be marked. You MUST attach the rough work sheet. MUST write proper SI units):

- Maximum nominal size of aggregate = 20mm — (1)
- Maximum water-cement ratio = 0.5 — (1)
- Minimum-cement content = 300 kg/m³ — (1)
- Exposure condition = Moderate — (1)
- Characteristic compressive strength = 45 MPa — (0.5)
- Standard deviation = 2.96 MPa — (2)
- Target mean strength = 49.88 MPa — (1)
- Approximate air content = 1% — (0.5)
- Free water-cement ratio = 0.39 — (1)
- Water content (net mixing) = 145 kg/m³ — (2)
- Cementitious material content = 409 kg/m³ — (1)
- Free water-cementitious material ratio = 0.35 — (0.5)
- GGBS content = 123 kg/m³ — (1)
- OPC content = 286 kg/m³ — (1)
- Volume of coarse aggregate = 0.4723 m³ — (4)
- Volume of fine aggregate = 0.2326 m³ — (4)
- Fine aggregate (SSD) = 605 kg/m³ — (0.5)
- Coarse aggregate (SSD) = 1181 kg/m³ — (1.5)
- Chemical admixture = 8.18 kg/m³ — (0.5)
- Water (to be added actually) = 148 kg/m³ — (1)
- Fine aggregate (actual) = 608 kg/m³ — (2)
- Coarse aggregate (actual) = 1175 kg/m³ — (2)